

AI for Everyone: Using a Neural Network to Identify Handwritten Digits

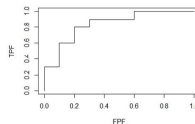
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2026 Hofstra Teaching and Learning Conference

January 21, 2026



**Mathematics of
Medical Imaging
(MoMI)
at Hofstra University**

Why am I here? Hofstra Cluster Hire + Data Science and AI (DSAI) Institute



DSAI Lunch and Learns + BS in Data Science and Machine Learning



About Docs Apply Announcements Policies



Lunch and Learn: Computational Skills Series

Join Us for a Fun Learning Experience!

Discover essential computational skills that are applicable across disciplines (i.e., suitable for both liberal arts and engineering students).

Topics will include data science and AI, Linux and Bash, VS Code Python, and high-performance computing (using the Star HPC cluster).

Event Details

Hosted once a month on Wednesdays,
starting September 24, 1-2:25 p.m. (Common Hour)

Location: Science and Innovation Center, Room 200

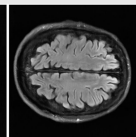
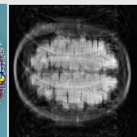
Food and coffee will be provided.

The schedule for the fall 2025 semester is as follows:

- **September 24: Bash vs. Genomics**
by Prof. Katherine Taylor (Biology)
- **October 15: VS Code and Python vs. Time Series**
by Prof. Paul Yang (Physics)
- **November 5: PyTorch vs. ML Handwritten Digits**
by Prof. Angel Pineda (Mathematics)
- **December 3: High-Performance Computing With GPUs**
by Prof. Michael Khayutin (Computer Science)

WATCH PREVIOUS SESSIONS ON YOUTUBE

Dr. Yuhao "Paul" Yang
Department of Physics and Astronomy



First Course in Data Science and Machine Learning Major: Foundations of Data Science



Scan for course description

Course number **CSC108 or MATH80**

Prerequisite CSC015; Co-requisite MATH071

Times: MW 9:40 am – 11:05 am

W 2:40 pm – 4:35 pm

Room: SIC 0126

Instructors

Dr. Angel Pineda, Mathematics
angel.pineda@hofstra.edu

Background: Magnetic Resonance Imaging (MRI)

- MRI uses a large magnetic field to image the magnetic properties of atomic nuclei (mostly hydrogen)
- There are no known negative effects for most types of scans



Background: Fourier Transform

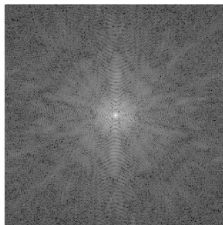
- MRI data to be modeled as the Fourier transform of the object:

$$F(k_x, k_y) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) e^{-2\pi i (xk_x + yk_y)} dx dy$$

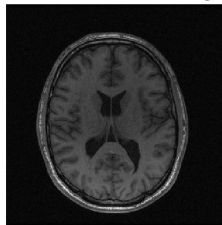
- The reconstruction process is the inverse Fourier transform:

$$f(x, y) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} F(k_x, k_y) e^{+2\pi i (xk_x + yk_y)} dk_x dk_y$$

MRI Data



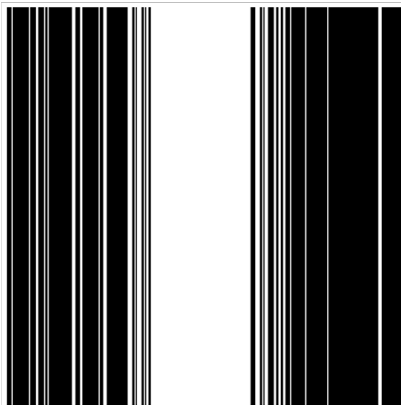
MRI Reconstructed Image



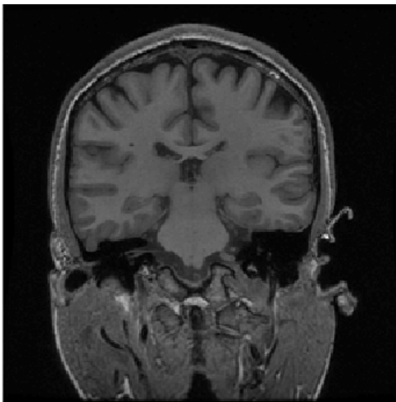
Background: Under-sampling

- 100 lines of 256 were sampled using the following mask:

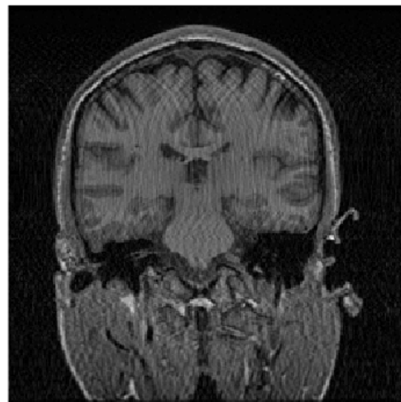
Fourier Sampling Mask



Fully Sampled Image

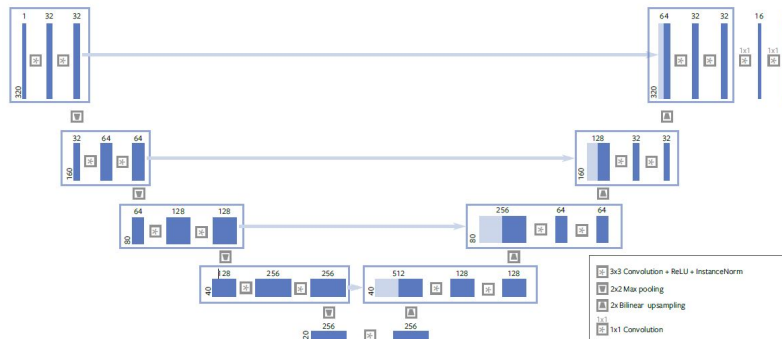


Under-sampled Image



Background: Neural Network Reconstruction in MRI

• U-Net Benchmark for Neural Network Reconstructions¹

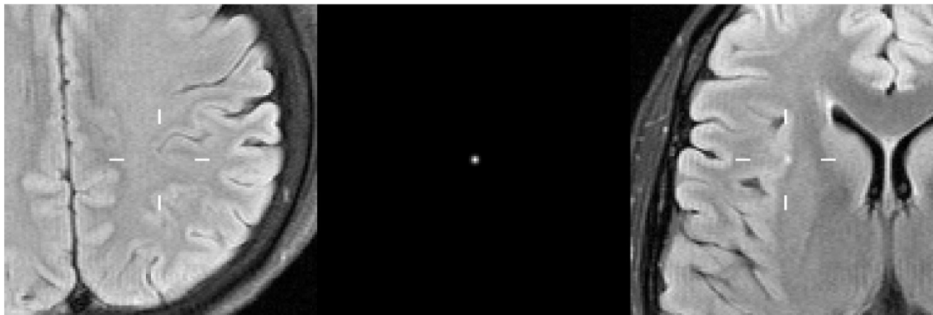


Single-coil U-Net baseline applied to validation data								
Acceleration	Channels	#Params	NMSE		PSNR		SSIM	
			PD	PDFS	PD	PDFS	PD	PDFS
4-fold	32	3.35M	0.0161	0.0531	33.78	29.90	0.81	0.631
	64	13.39M	0.0157	0.0528	33.90	29.9	0.813	0.633
	128	53.54M	0.0154	0.0525	34.01	29.95	0.815	0.634
	256	214.16M	0.0154	0.0525	34.00	29.95	0.815	0.636

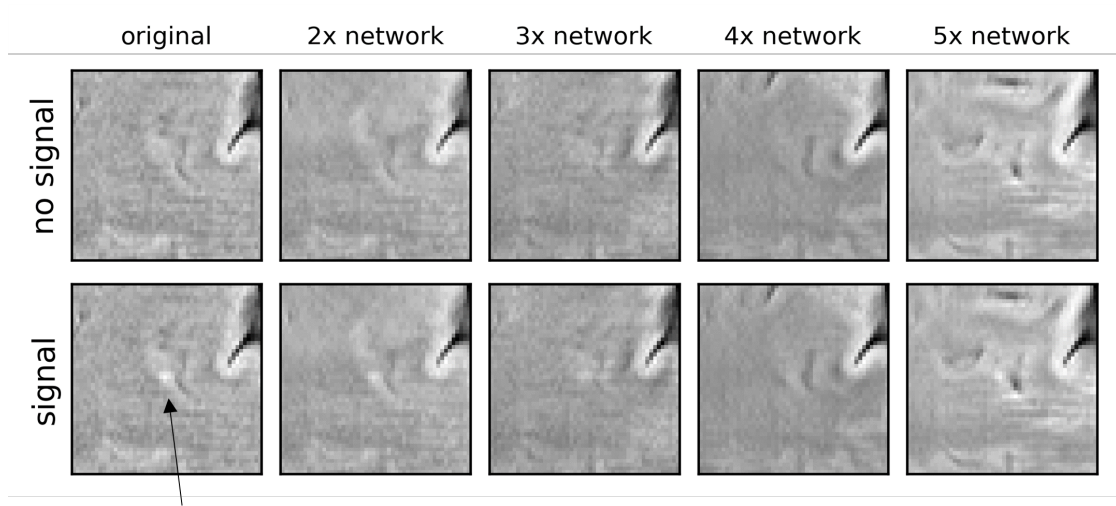
¹Zbontar, Knoll et. al., fast MRI: An Open Dataset and Benchmarks for Accelerated MRI, *arXiv:1811.08839*

Psychophysical Studies: 2 Alternative Forced Choice (2-AFC)

- Three images
 - anatomical background with signal
 - isolated signal
 - anatomical background without signal
- 200 image pairs per experiment
- Output: percentage chosen correctly



How much to undersample with a neural network?



- Introduce task based evaluation of network reconstruction

Current Research

- Statistical Experimental Design for Psychophysical Studies
- Neural Networks to Predict Human Observer Performance

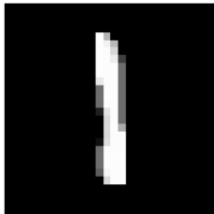


Modified National Institute of Standards and Technology (MNIST) Dataset

GT: 7 | Pred: 3



GT: 1 | Pred: 1



GT: 5 | Pred: 5



GT: 7 | Pred: 7



GT: 9 | Pred: 9



GT: 3 | Pred: 3



History of the MNIST Dataset and Neural Networks

- Original images (NIST) were of varying sizes and quality; MNIST standardized them to 28×28 pixels, greyscale, centered the digit.
- Served as a benchmark for early machine learning development.
- Became too easy with the advent of neural networks
- Hinton's 2012 paper on Imagenet (harder than MNIST) using a network with 60 million parameters has 189000 citations!

ImageNet Classification with Deep Convolutional Neural Networks

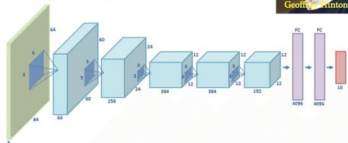
Alex Krizhevsky
University of Toronto
alexkr@cs.toronto.edu

Ilya Sutskever
University of Toronto
ilya@cs.toronto.edu

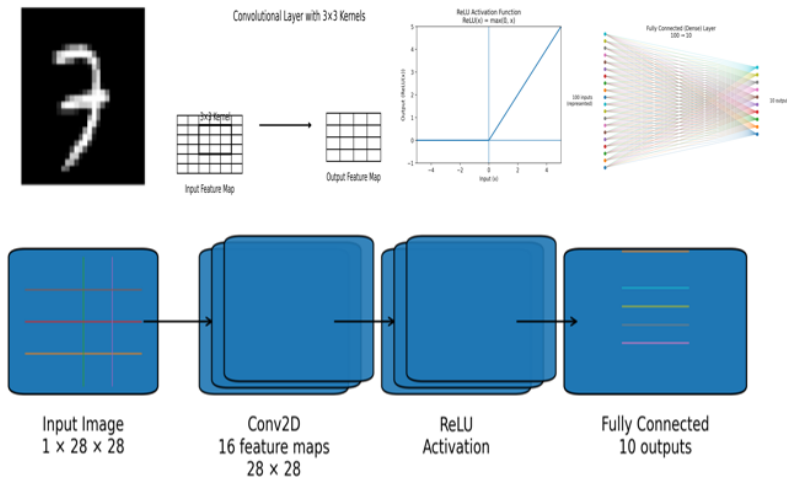
Geoffrey E. Hinton
University of Toronto
h1@mtl.utoronto.ca

Abstract

We trained a large, deep convolutional neural network to classify the 1.2 million high-resolution images in the ImageNet ILSVRC-2010 contest into the 1000 different classes. On the test data, we achieved top-1 and top-5 error rates of 37.5% and 17.0% which is considerably better than the previous state-of-the-art. The second network, which has 60 million parameters and 650,000 neurons, consists of five convolutional layers, some of which are followed by max-over-time layers, and three fully-connected layers with a final 1000-way softmax. To make training faster, we used non-saturating neurons and a very efficient CPU implementation of the convolution operation. To reduce overfitting in the fully-connected layers we employed a recently-developed regularization method called "dropout" that proved to be very effective. We also entered a variant of this model in the ILSVRC-2012 competition and achieved a winning top-5 test error rate of 15.3%, compared to 26.2% achieved by the second-best entry.



Our simple convolutional neural network



Per-layer parameter counts

- Conv2d($1 \rightarrow 16$, 3×3 , bias=True)
Parameters = $16 \times (1 \times 3 \times 3) + 16$ (biases) = 160
- Linear($12,544 \rightarrow 10$, bias=True)
Parameters = $10 \times 12,544 + 10$ (biases) = 125,450

Total trainable parameters

$$160 + 125,450 = 125,610$$

Using AI (Hofstra Chat GPT) to prepare this talk

ChatGPT 5 ▾

Create code for a jupyter notebook that classifies the MNIST data using pytorch

Thought for 5s >

Nice — here's a complete Jupyter Notebook you can copy/paste into a `.ipynb` (or into notebook cells) that trains a simple CNN on MNIST using PyTorch. It includes dataset download, model, training & validation loops, accuracy/loss plots, and model saving. It is designed to run on CPU or GPU automatically.

I keep things clear and commented so you can modify hyperparameters easily. Ready? — paste the following into notebook cells in order.

MNIST classification with PyTorch — Notebook content

markdown

 Copy code

```
# MNIST classification with PyTorch
```



Star HPC

Search docs...

GETTING STARTED

1. About Star

Overview

Hardware

Login Node

Compute Nodes

Storage System

Our vision

Operations Team

The Board

2. Quick Start Guide

GETTING HELP

1. Frequently asked questions

2. How to write good support requests

3. Contact Us

JOBS

1. Overview

2. Submitting Jobs

3. Job Scheduling

4. Monitoring Jobs

SOFTWARE

1. Environment modules

Star HPC

Hardware

Login Node

- IBM System x3550 with 128GB RAM
- DL325 Gen10+ with an 8-core EPYC processor, 128GB RAM
- DL385 Gen10+ v2 with 2x AMD 32-core EPYC processors, 256GB RAM

Compute Nodes

HPE Apollo 6500 Gen10

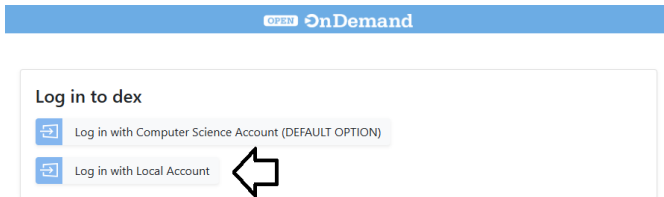
Attribute\Node Name	gpu1 and gpu2
Model Name	HPE ProLiant XL675d Gen10 Plus; Apollo 6500 Gen10 Plus Chassis
Processors	AMD EPYC 7513
Sockets	2
Cores per Socket	32
Threads per Core	2
Memory	1024 GiB Total Memory (16 x 64GiB DIMM DDR4)
GPU	8 SXM NVIDIA A100s
Local Storage (Scratch space)	6.4TB (5.8TiB) SSD

HPE DL385 Gen10

Attribute\Node Name	cn01
Model Name	HPE ProLiant DL385 Gen10 Plus v2
Processors	AMD EPYC 7513 32-Core Processor
Sockets	2
Cores per Socket	32
Threads per Core	2
Memory	256GiB Total Memory (16 x 16GiB DIMM DDR4)

Using the STAR HPC Cluster to recognize handwritten digits Part I

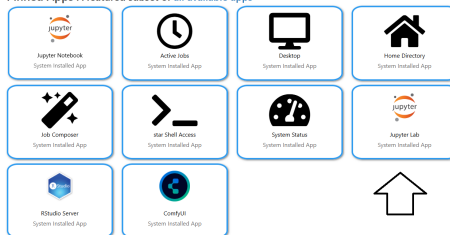
- start a session at <https://ood.star.hofstra.edu/> using your temporary login account
- login to Open onDemand with a local account



- start a Jupyter Lab session

OnDemand provides an integrated, single access point for all of your HPC resources.

Pinned Apps A featured subset of all available apps



Using the STAR HPC Cluster to recognize handwritten digits Part II

- choose QoS normal, GPU A100, 1 core, 1 GPU, 1 hour and Launch

Interactive Apps

Desktops

Desktop

Development

ComfyUI

Servers

Jupyter Lab

Jupyter Notebook

RStudio Server

Jupyter Lab

This app will launch a Jupyter Lab server on one or more nodes.

Account

lunch_and_learn

Partition

defq

QoS

normal

Node type

GPU A100

- **Standard Compute**
- **GPU Enabled**
These are HPC machines with GPUs.
- **Large Memory**
These are HPC machines with very large amounts of memory.

Number of cores

1

Number of cores on node type. Minimums and maximums will change depending on the choice of node type and cluster.

Gpus

1

Number of hours

1

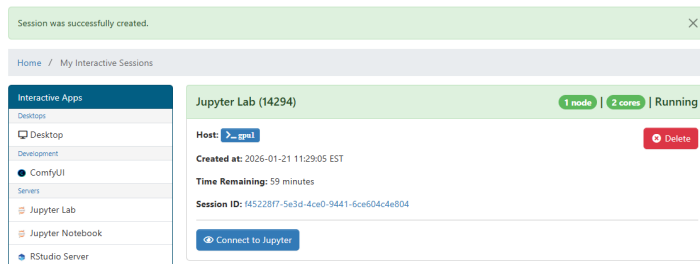
☐ I would like to receive an email when the session starts

Launch

* The Jupyter Lab session data for this session can be accessed under the [data root](#) directory.

Using the STAR HPC Cluster to recognize handwritten digits Part III

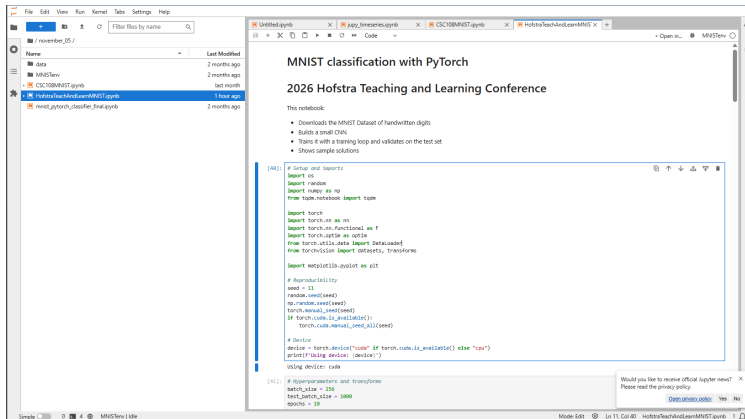
- Wait patiently for a few seconds until you see a Connect to Jupyter blue button...
- Click on the Connect to Jupyter button



- Click on november_05 folder if not there already
- open the HofstraTeachAndLearnMNIST Jupyter notebook
- wait until we are all there :-)

Using the STAR HPC Cluster to recognize handwritten digits Part IV

- Click on november_05 folder if not there already
- open the HofstraTeachAndLearnMNIST Jupyter lab



- wait until we are all there :-)